

Course Path

STAT 109 — Summer Session 1, 2026 (online, async)

This page shows the recommended path through the course during the five-week summer session from June 1 to July 2, 2026. You may work at your own pace within each module window, but the modules should be done in the order shown. Check [Canvas](#) to turn in work, take quizzes, see grades, and join discussions.

Each project module follows the same pattern: **Read 0** (project preview) → read → update your reference sheets → low-stakes quizzes (as many times as you want) → lab (Wednesday) → synthesis quiz (Thursday) → project milestone (Friday) → project (Sunday). Module 5 is shorter and ends on Thursday, July 2.

Reference sheets ([methods map](#), [R reference](#)) grow a little in every module 1–5. All readings also live on the [lectures](#) index if you want the full list.

Outcomes below are tagged **essential** (needed for the module project or core module goals) or **optional** (helpful enrichment or stretch).

At a glance

Module	Dates	Project due
0 — Welcome	Open before Jun 1	—
1 — Testing a Claim	Jun 1–7	Sun Jun 7
2 — Estimating the Truth	Jun 8–14	Sun Jun 14
3 — Discovering Patterns	Jun 15–21	Sun Jun 21
4 — Designing Fair Comparisons	Jun 22–28	Sun Jun 28
5 — Looking Ahead	Jun 29–Jul 2	Thu Jul 2

Module 0 — Start here

Complete this before Module 1

Step	Materials	Learning outcomes
Read	Syllabus , course path , resources	Essential: Know where readings, labs, and projects live vs what happens in Canvas. Essential: Understand grading, due-date pattern, and academic honesty / AI expectations (syllabus). Optional: Full policy details you will not need until later in the term.
Reference sheets	Methods map , R reference	Essential: Set up two places to create your two references over the next 5 weeks. You will update these each module and can use them on assignments. Optional: Skim the methods map layout and R reference before Module 1 labs.
Lab (Wed)	Demo lab , Exercise lab	Essential: Run R in Google Colab (save a copy, run cells). Essential: Work with vectors, logical TRUE/FALSE, and simple simulation. Optional: Estimating probabilities by simulation (helps later probability intuition, not required for Project 1).
Canvas	Module 0 discussion	Essential: Introduce yourself and confirm you can reach assignments and discussions.

Module 1 — Testing a Claim (Jun 1–7)

Is this observation surprising?

Lab Wed Jun 3 · Synthesis Thu Jun 4 · Milestone Fri Jun 5 · **Project 1 — Testing a Claim** Sun Jun 7 · Discussion: something surprising about Lumberjacks (Canvas)

Step	Materials	Learning outcomes
Read 0	Module 1 Read 0	Essential: Project preview — big question, sample ESP report (PDF) , what to notice in each section, link to Project 1 . ~5 min before Read 1.
Read	Module 1 Read 1	Essential: Probability language through binomial pmf — sample spaces, rules, counting, and the binomial process.
Quizzes (Def & Notation)	Canvas Module 1 — Quiz A	Essential: Notation, sample space, addition rule, conditional probability, independence. Open notes; unlimited attempts.
Read more	Module 1 Read 2	Essential: Binomial test workflow and R — hypotheses, null plots, p-values.
Quizzes (Concept Check)	Canvas Module 1 — Quiz B	Essential: Counting, conditional applications, combinations, BRP identification. Open notes; unlimited attempts.
Reference sheets	Methods, R reference	Essential: Probability vocabulary (sample space through independence) on statistics sheet. Essential: One-proportion / binomial test row on methods map. Essential: R patterns for <code>dbinom</code> , <code>pbinom</code> , <code>geom_col</code> , <code>geom_vline</code> .
Lab (Wed)	Demo lab, Conditional demo, Exercise lab, Project 1 milestone	Essential (demos): simulation review, data structures, conditional simulation, binomial R tools, null plot. Essential (exercise): combined practice notebook (Parts I–III). Essential (milestone): Project 1 precheck. Update R reference .
Synthesis quiz (Thu)	Canvas Module 1 — Synthesis	Essential: Full binomial test workflow, Module 1 R practice, interpretation items. After labs. Two attempts; open notes.
Milestone (Fri)	Project 1 milestone	Essential: Phase A — question, p , H_0/H_a , null graph before data collection.
Project (Sun)	Project 1 — Testing a Claim	See project outcomes table below.

Project 1 — Testing a Claim

Outcome	Tag
Choose a real binomial process ($n = 25$) that is not a class example; defend all four BRP conditions in your own context	Essential
State H_0 and H_a for population proportion p with a numeric null value	Essential
Report x , n , $\hat{p}$; descriptive bar/pie chart in R	Essential
Plot null binomial distribution with <code>dbinom</code> + <code>geom_col</code> ; mark observed x	Essential
Compute p-value with <code>pbinom</code> matching alternative direction; interpret in one sentence	Essential

Outcome	Tag
One-page report with Introduction / Data Collection / Data Analysis / Conclusion	Essential
Use the Read 2 ESP worksheet as a model without copying ESP as your study	Essential

Module 2 — Estimating the Truth (Jun 8–14)

Building high-quality estimates from samples.

Lab Wed Jun 10 · Synthesis Thu Jun 11 · Milestone Fri Jun 12 · **Project 2 — Estimating the Truth** Sun Jun 14 · Discussion: a plant you appreciate (Canvas)

Step	Materials	Learning outcomes
Read 0	Module 2 Read 0	Essential: Project preview — finite population, SRS, 95% CI focus; tour sample report (PDF) . ~5 min before Read 1.
Read	Module 2 Read 1	Essential: Binomial $\mu = np$, $\sigma = \sqrt{np(1-p)}$; normal approximation & Empirical Rule; sampling distribution & 95% CI (proportion and mean); sampling frame, bias, precision, SRS and design types.
Quizzes (Def & Notation)	Canvas Module 2 — Quiz A	Essential: Vocabulary for estimation and sampling (frame, SRS, bias, CI language). Open notes; unlimited attempts.
Read more	Module 2 Read 2	Essential: z-scores & standard normal; methods-map habit; mean/median, histogram, dot plot in tidyverse. Optional: <code>prop.test()</code> / one-proportion z-test; one-sample <code>t.test()</code> (Project 2 uses CI only).
Quizzes (Concept Check)	Canvas Module 2 — Quiz B	Essential: z-scores, toolbox organization, one-variable summaries. Open notes; unlimited attempts.
Reference sheets	Methods, R reference	Essential: One-mean and one-proportion CI rows; FPC note for finite populations. Essential: z-score row before toolbox section.
Lab (Wed)	Module 2 demo, Exercise lab	Essential: Choose among CI situations (mean vs proportion; finite N ; FPC). Essential: Check conditions and compute 95% CI in R.
Synthesis quiz (Thu)	Canvas Module 2 — Synthesis	Essential: CI computation and interpretation; SRS vs convenience; FPC decision; exercise-lab R patterns; tidyverse summaries. After labs.
Milestone (Fri)	Canvas Module 2	Essential: Named finite population (N), variable type, planned SRS method, and which CI type you will use.
Project (Sun)	Project 2 — Estimating the Truth	See project outcomes table below.

Project 2 — Estimating the Truth

Outcome	Tag
Define a listable finite population (N) and one variable (binary $\rightarrow$ proportion, or numerical $\rightarrow$ mean)	Essential
Draw an SRS of size n from $\{1, \dots, N\}$ without replacement; collect data	Essential
Compute point estimate ($\hat{p}$ or $\bar{x}$)	Essential
Choose correct 95% CI (mean t -interval or proportion interval; apply FPC when N is finite)	Essential
State and check interval conditions in context	Essential
Interpret CI with the “I am 95% confident that...” sentence	Essential
Submit one-page summary + runnable R notebook (mean or proportion template)	Essential
Sample size calculation / professional power analysis	Optional
Hypothesis test about the parameter (not required by rubric)	Optional

Module 3 — Discovering Patterns (Jun 15–21)

Describing a dataset with graphs and summaries.

Lab Wed Jun 17 · Synthesis Thu Jun 18 · Milestone Fri Jun 19 · **Project 3 — Discovering Patterns** Sun Jun 21 · Discussion: a place from home (Canvas)

Step	Materials	Learning outcomes
Read	Lecture 25 , Lecture 26 , Lecture 27	Essential (L26): Scatterplots; describe form, direction, strength. Essential (L27): Two-way tables; joint/row/column proportions; stacked bar charts. Essential (L25): Compare numerical outcome across two groups with summaries, boxplots, <code>group_by</code> + <code>summarize</code> . Optional (L25): Welch two-sample t -test and inferential group comparison (Project 3 is descriptive only).
Reference sheets	Methods , R reference	Essential: Describe rows for one categorical, one numerical, two categorical, two numerical, and numerical-by-group. Essential: <code>ggplot</code> patterns for histogram, boxplot, bar, scatter, stacked bars. Optional: Inferential rows for two-group tests (not required for Project 3).
Quizzes	Canvas Module 3 — Def & Notation + Concept Check (Quiz 9)	Essential: Classify numerical vs categorical; tidyverse verbs (<code>filter</code> , <code>group_by</code> , <code>summarize</code> , <code>count</code>). Essential: Mean/median/SD/IQR language (Quiz 9 / L24). Optional (review, Module 2 L22): z-scores and standard normal probabilities. Optional: One-proportion z-test / <code>prop.test()</code> (Quiz 9 / L23).
Lab (Wed)	Lab 9 , Lab 10	Essential (Lab 9): Load data; recode factors; univariate summaries and plots; <code>filter</code> and grouped summaries. Essential (Lab 10): Missing data; <code>read_csv</code> from URL or upload; bivariate plots and tables (scatter, boxplot by group, two-way counts/proportions).

Step	Materials	Learning outcomes
Synthesis quiz (Thu)	Canvas Module 3 (Quiz 10)	Essential: Read graphs; pick appropriate summaries by variable type; interpret two-way tables and stacked bars. Optional: Inferential reading of group-comparison plots if included.
Milestone (Fri)	Canvas Module 3	Essential: Chosen dataset link, four variables with types, and plan for two figures + four numeric facts.
Project (Sun)	Project 3 — Discovering Patterns	See project outcomes table below.

Project 3 — Discovering Patterns

Outcome	Tag
Choose a dataset (TidyTuesday or other) with 4 variables and 10 rows; document source	Essential
Classify four variables as numerical or categorical	Essential
Univariate summaries and plots for each of the four variables	Essential
Bivariate exploration for variable pairs (tables/plots appropriate to types)	Essential
One-page PDF: context, two best graphics, 4 labeled numeric summaries, four findings	Essential
Notebook runs top-to-bottom with all code used (4 univariate + 6 bivariate plots in notebook; only 2 plots on one-pager)	Essential
Part A short answers in Canvas	Essential
Inferential tests or population generalization beyond the dataset	Optional (explicitly out of scope — descriptive statistics only)
Your own research question beyond exploratory findings	Optional

Module 4 — Designing Fair Comparisons (Jun 22–28)

Comparing two groups responsibly.

Lab Wed Jun 24 · Synthesis Thu Jun 25 · Milestone Fri Jun 26 · [Project 4 — Designing Fair Comparisons](#) Sun Jun 28 · Discussion: a place in Humboldt (Canvas)

Step	Materials	Learning outcomes
Read	Lecture 28 , Lecture 29 , Lecture 30	Essential (L28): EV vs RV; observational studies vs experiments; confounding; DAG with one confounder; “by chance” vs causal language. Essential (L29): RCT definition; what random assignment buys; interpreting group differences under randomization. Essential (L30): Pick two-group test or CI from methods map by variable types; worked examples for mean difference or proportion difference. Optional (L28): Adjustment sets / backdoor criterion beyond one confounder. Optional (L29): When RCT is impossible.

Step	Materials	Learning outcomes
Reference sheets	Methods, R reference	Essential: Two-group inferential rows (Welch t , two-proportion / χ^2 as appropriate). Essential: Study-design vocabulary on statistics sheet. Essential: R pipeline: <code>group_by</code> , plots by group, <code>t.test</code> / <code>prop.test</code> / <code>chisq.test</code> .
Quizzes	Canvas Module 4 — Def & Notation + Concept Check (Quiz 11)	Essential: RCT vs observational; what a p-value does and does not show; confounding and causal wording (Quiz 11 / L28–29). Essential: Match design to research question.
Lab (Wed)	Lab 11	Essential: Simulate a population with known truth; draw a sample; run the test that matches variable types. Essential: Validate code and method choice against known parameters. Essential: <code>set.seed()</code> , tidy simulation, group summaries, appropriate test.
Synthesis quiz (Thu)	Canvas Module 4 (Quiz 11)	Essential: Design + methods table integration; choose test for two-group comparison; interpret p-value and CI in context. Optional: ANOVA / χ^2 independence beyond two-group focus (Quiz 12 topics — not on Module 4 read list).
Milestone (Fri)	Canvas Module 4	Essential: Assigned research question; draft answers to Part 1 design questions; plan for simulated data structure.
Project (Sun)	Project 4 — Designing Fair Comparisons	See project outcomes table below.

Project 4 — Designing Fair Comparisons

Outcome	Tag
Answer ten study-design questions (EV/RV, operational definitions, hypotheses, RCT vs observational, confounder + DAG, test name from methods table)	Essential
Simulate reproducible dataset (<code>set.seed</code>) with binary EV, response, and confounder when needed	Essential
Descriptive summaries and plots by group (numerical → boxplot/summaries; binary → stacked bars / proportions)	Essential
Run appropriate two-group test / CI; report p-value and interval; discuss “by chance” explanation	Essential
One-page summary written for the classmate who asked the question	Essential
Runnable notebook with simulation + analysis sections	Essential
Real-world data collection for the assigned question	Optional (project uses simulation)
Adjust for confounding in the inferential model	Optional (confounder included in design/DAG and plots; test does not remove confounding)

Module 5 — Looking Ahead (Jun 29–Jul 2)

Using data to make predictions.

This module is four days. Lab Wed Jul 1 · **Project 5 — Looking Ahead**, synthesis quiz, milestone, and portfolio package due Thu Jul 2 · Discussion: predicting a future observation (Canvas)

Step	Materials	Learning outcomes
Read	Lecture 35, Lecture 36	Essential (L35): Pearson r for linear form; least-squares line; residuals; train/test and R^2 intuition. Essential (L36): Inference for slope; CI and prediction interval from <code>predict(..., interval = "prediction")</code> ; interpret slope/intercept in context. Optional (L35): Model comparison beyond one predictor.
Read more	Lecture 37	Optional: Logistic regression for binary response (<code>glm</code> , probabilities) — not required for Project 5 options (linear regression suggested).
Reference sheets	Methods, R reference	Essential: Two-numerical describe + regression / prediction rows. Essential: <code>lm</code> , <code>predict</code> , scatterplot with fitted line on R sheet. Optional: Binary RV + logistic row (L37).
Quizzes	Canvas Module 5 — Def & Notation + Concept Check (Quiz 12)	Optional: Paired vs independent designs; ANOVA / χ^2 concept items (Quiz 12 / L31–33 — not on Module 5 read list). Essential: Regression vocabulary as selected in your Canvas bank.
Lab (Wed)	Lab 12, Lab 13	Essential (Lab 12): Scatterplot description; <code>cor</code> ; fit <code>lm</code> ; prediction and intervals. Optional (Lab 12): Logistic <code>glm</code> demo. Essential (Lab 13): Google Sheet → CSV → R; team workflow for Project 5.
Synthesis quiz (Thu)	Canvas Module 5 (Quiz 13)	Essential: Read scatterplots; interpret r , slope, intercept; prediction vs confidence interval; read <code>predict</code> output (Quiz 13 / L35–36).
Milestone (Thu)	Canvas Module 5	Essential: Team, option (nesting dolls or stones), variables defined, data collection underway.
Project (Thu)	Project 5 — Looking Ahead	See project outcomes table below.

Project 5 — Looking Ahead

Outcome	Tag
Form a team (1–4); choose Option A (nesting dolls) or B (stone container)	Essential
Define measurement rules and collect team dataset (CSV / shared sheet)	Essential
Fit at least one simple linear model <code>lm(response ~ predictor)</code>	Essential
Report prediction equation, plugged prediction for target case, 95% prediction interval	Essential
One-page PDF: intro, methods, results, conclusion, team contributions	Essential
Runnable team <code>.ipynb</code> from data load through model and prediction	Essential
Compare multiple models / predictors and choose best	Optional (encouraged but one SLR minimum)
Logistic regression for binary outcomes	Optional (L37 / Lab 12 demo)
Portfolio package: both reference sheets + all five projects polished for submission	Essential

Notes

- Modules unlock in order in Canvas ($0 \rightarrow 1 \rightarrow \dots \rightarrow 5$).
- Quizzes are open notes: use your reference sheets and the methods map.
- Definitions & Notation and Concept Check: unlimited attempts. Synthesis: two attempts (highest score counts).
- The milestone is an ungraded checkpoint — it helps you and me catch problems before the project is due.
- Modules 1–4: lab Wednesday, synthesis Thursday, milestone Friday, project Sunday.
- Module 5: lab Wednesday July 1; synthesis, milestone, project, and portfolio package due Thursday July 2.
- Quiz source files (`quizzes/quiz-*.qmd`) are practice banks; Canvas quizzes may use a subset. Items marked **optional** above may still appear — they are not required for the module project.
- If dates shift when the term opens, Canvas is the official calendar; this page will be updated to match.